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Introduction to Git and GitHub

Why we need to use version control

为什么我们需要版本控制





Why we need to use version control

Why we need to use version control

fast creative leader focus transpiration bold inspiration



Assume...

You are writing an plan.



Draft.md



1st_version.md



2st_version.md



final_ver.md



latest_ver.md



real_latest_ver.md



Acturally_final.md

Assume...

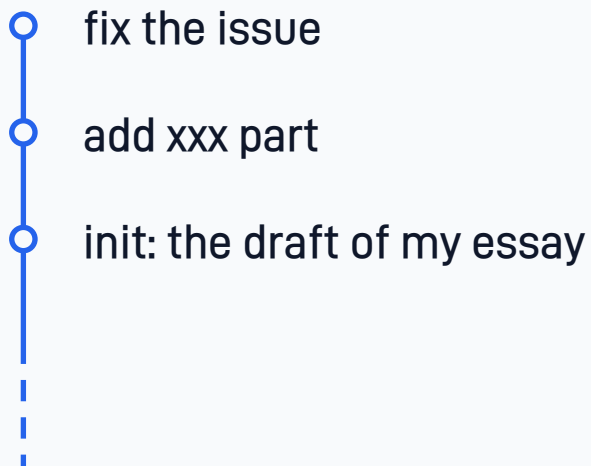
You are writing an plan.

In fact, version control tools can automatically help you manage these versions.

事实上，版本控制工具就可以实现自动帮你管理这些版本。



My_plan.md





Version Control System

A version control system (VCS) tracks the history of changes made by people and teams as they collaborate on a project. When developers make changes to a project, they can restore any earlier version of the project at any time.

版本控制系统（VCS）跟踪人员和团队在项目上进行协作时的更改历史记录。当开发人员对项目进行更改时，可以随时恢复项目的任何早期版本。

查看项目历史记录以找出：

What	/	Who	/	When	/	Why	edit the file(s)
进行了哪些更改		谁进行了修改		何时进行了更改		为什么进行修改	

In a distributed version control system, every developer has a complete copy of the project and its history.

在分布式版本控制系统中，每个开发人员都有项目和项目历史记录完整副本。



Git

The most popular version control system.

Nearly all developer use it for development.

Just like the time machine for your project.

What can it do?

- Record the file history. 记录文件修改历史
- Corporate development for projects. 协作开发项目
- Back to any version if you want. 回退到任意版本

Core concept of Git

Git 的四大法宝



Repository
仓库



Commit
提交



Branch
分支



Merge
合并





Core concept of Git

Git 的四大法宝



Repository
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Branch
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Merge
合并

Repository:

Save project's file and history. 保存文件和历史记录

Commit:

Record a edit snapshot. 记录一次修改快照

Branch:

A independent development path. 独立的开发路线

Merge:

Combine multi-branches. 合并多个分支



How git work?

Git 是如何工作的

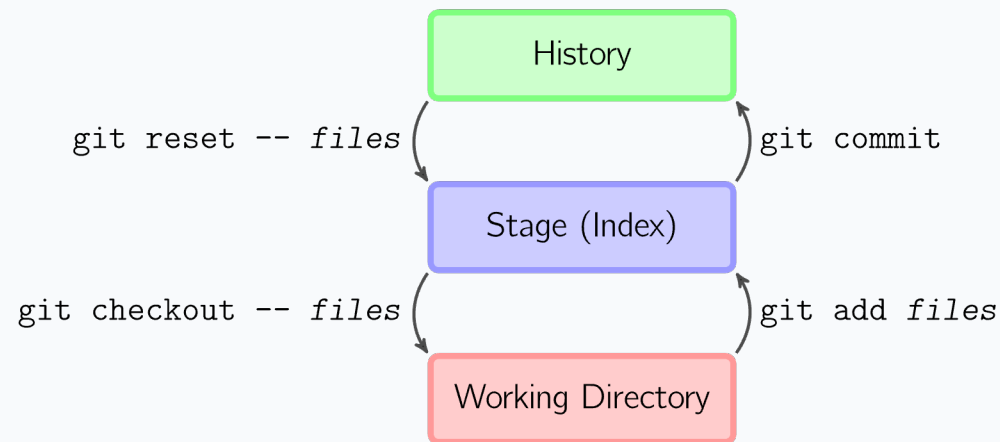
A trip for a file:

工作区(Working Directory) → 暂存区(Staging Area) → 仓库(Repository)

What you need to do:

修改 edit → 暂存 stage → 提交 commit

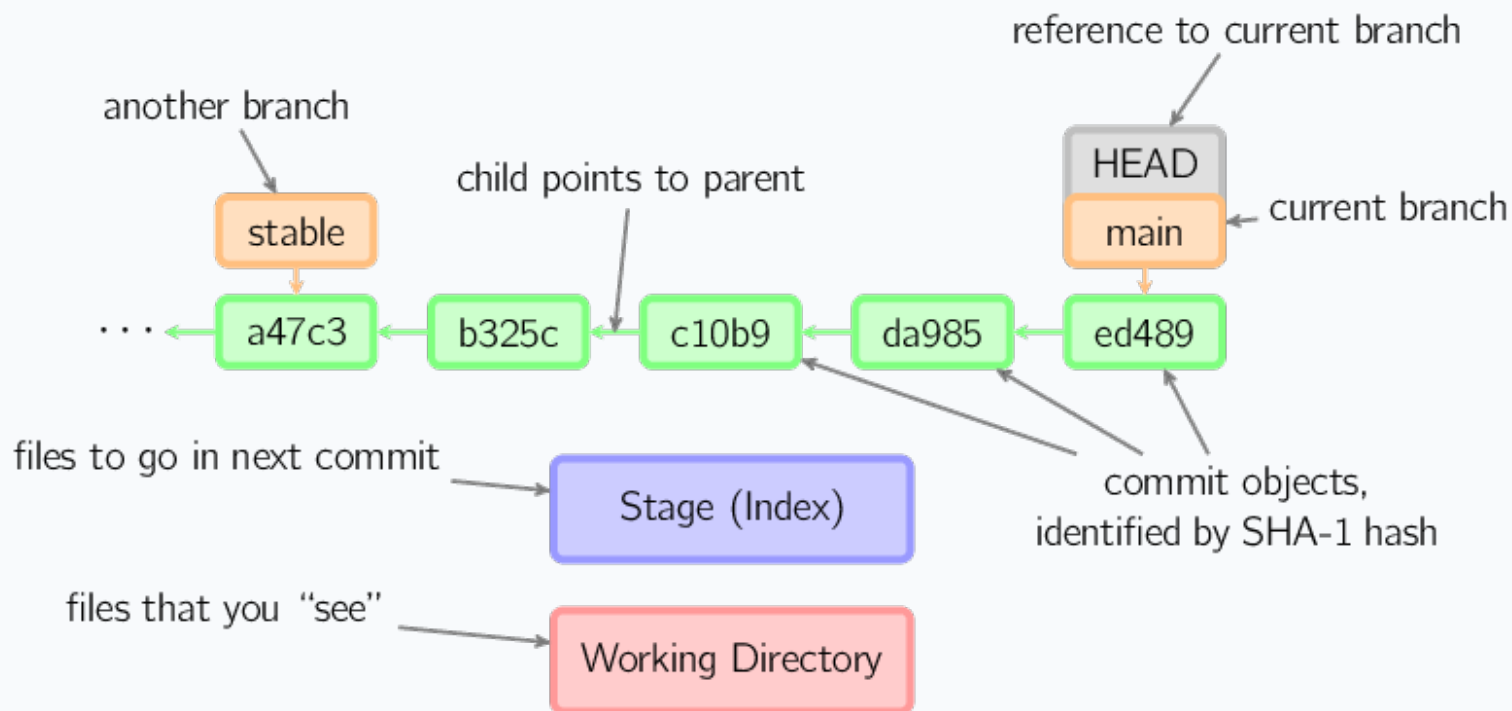
git add **git commit**





How git work?

Git 是如何工作的





GitHub

Official GitHub Website 全球唯一正版GitHub网站: <https://github.com/>

GitHub is a cloud-based platform where you can store, share, and work together with others to write code. GitHub 是一种基于云的平台，可在其中存储、共享并与他人一起编写代码。

Storing your code in a "repository" on GitHub allows you to:

- **Save your work to cloud.**
- **Showcase or share** your work.
- **Track and manage** changes to your code over time.
- Let others review your code, and make suggestions to improve it.
- **Collaborate** on a shared project, without worrying that your changes will impact the work of your collaborators before you're ready to integrate them.

Collaborative working, one of GitHub's fundamental features, is made possible by the open-source software, Git, upon which GitHub is built.



Useful git command

常用 Git 命令

`git init`

`git add .`

`git commit -m "description"`

`git pull`

`git push`

`git status`

`git log`

`git diff`

Useful reference:

<https://docs.github.com/zh/get-started>

<https://rogerdudler.github.io/git-guide/index.zh.html>



Connect your repo to GitHub

```
git remote add origin <url>    # 连接远程仓库  
git push -u origin main         # 推送本地代码  
git clone <url>                 # 从 GitHub 克隆项目
```



Team collaboration and branching

Team develop flow:

1. Fork repo
2. Clone to local
3. Create a new branch
4. Push , and then create Pull Request

Git-flow:

Git Flow is a **branching workflow model** for Git, introduced by Vincent Driessen in 2010.

It provides a structured way for **team collaboration** and **release management**, especially for medium to large projects with regular release cycles.

Git Flow defines a **standard branch structure** and rules for working on features, releases, and hotfixes.

Git Flow 是一种基于 Git 分支模型的 **开发流程规范**,

由 Vincent Driessen 在 2010 年提出, 旨在让多人协作开发更有条理、更安全。

Git Flow 定义了用于处理功能、发布和修补程序的标准分支结构和规则。



Git Flow

Branch rules:

There are 5 Core Branches in Git Flow

main - Stores the official production-ready code

develop - Contains the latest development changes

Permanent Branch – always in your repo

Temporary Branch - delete after use

feature/xxx - Used for developing new features

fix/xxx - Emergency fixes for production bugs

release/xxx (optional) - Prepares code for a new production release

Commit message rules:

<type>(<scope>): <subject>

Subject:

1. Small letter begin
2. Start with a verb
3. Briefly
4. Easy to read

Types:

- feat:
- fix:
- docs:
- test:
- chore:



Git Flow

Branch:

✗ vote

✓ feature/add-user-vote

✗ bugfix

✓ fix/xxx-bug

✗ Baicen's branch

✓ feature/baicen_add-login

Commit message:

✗ vote

✓ feat: add vote func

✗ feat: vote

✓ feat: add vote func

✗ feat: This is to add vote func of the system

✓ feature/baicen_add-login



GitHub Education

What is GitHub Education?

- A program by GitHub to help students and teachers learn, teach, and build software.
- Provides **free access to professional tools** for development, cloud, and productivity.
- Supports learning and collaborative projects in real-world software development.

Student Developer Pack Includes:

- Free **GitHub Pro** account
- Access to cloud services: **AWS, DigitalOcean, Heroku**
- Development tools: **Visual Studio Code, GitKraken, JetBrains IDEs**
- Learning resources: **DataCamp, Educative, Canva**, etc.
- Discounts on domain registration: **Namecheap**, etc.

How to Apply:

1. Go to [GitHub Education](#)
2. Sign in with your **GitHub account**
3. Verify **student status** (school email or student ID)
4. Submit the application and wait for approval
5. Once approved, enjoy the **free student benefits**



References

[1] Get started Github, GitHub. <https://docs.github.com/zh/get-started>

[2] Git reference, Git. <https://git-scm.com/docs>

[3] Git 简明指南, <https://rogerdudler.github.io/git-guide/index.zh.html>

[4] A Visual Git Reference, Mark Lodato., <https://marklodato.github.io/visual-git-guide/index-en.html>

For learning Git,

Practice more,

try more,

and ask more